

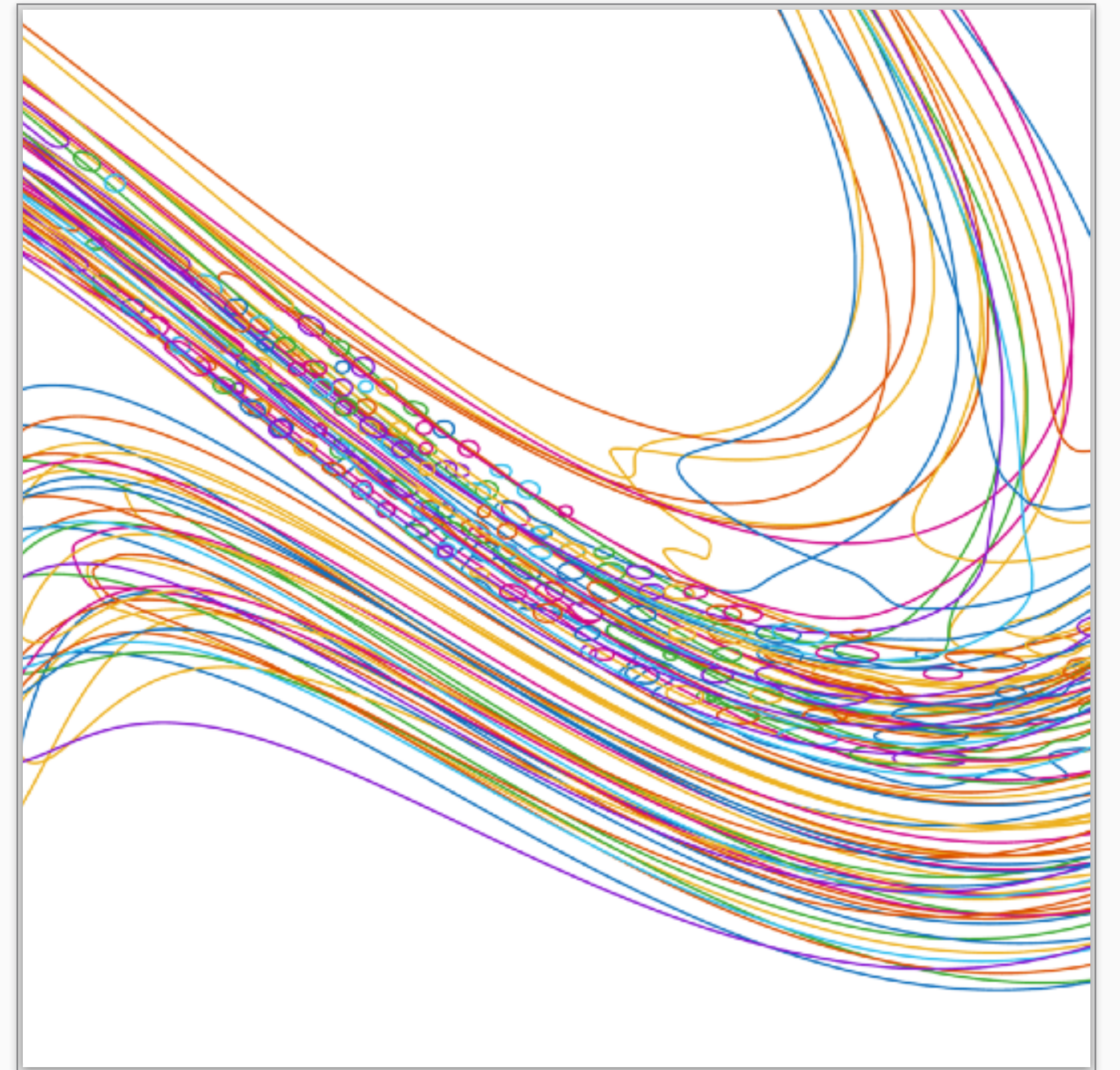
# MICROSCALE TOPOLOGICAL SURGERY FOR TRACKING FILAMENT BREAKUP

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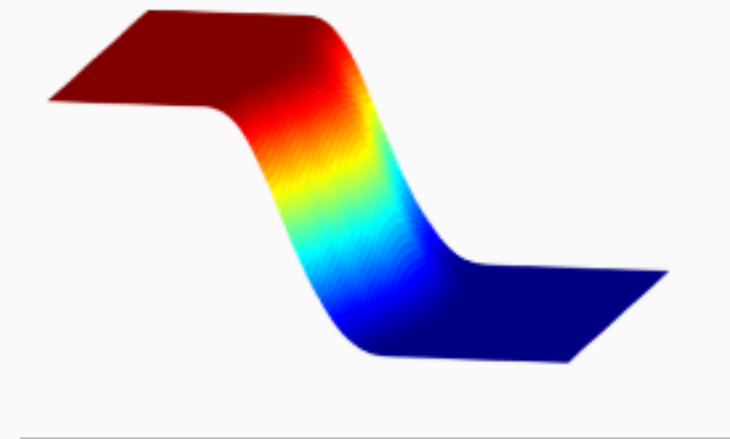
# EFFICIENT AND ACCURATE INTERFACE TRACKING

## EULERIAN CAPTURING

Interface is implicitly captured  
by a level-set function on a  
background mesh

$$\Gamma(t) = \{x \in \mathbb{R}^d : \phi(x, t) = 0\}$$

$$\partial_t \phi + u \cdot \nabla \phi = 0$$



## LAGRANGIAN TRACKING

Interface is explicitly tracked  
as a codimension—1  
parametrized surface

$$\Gamma(t) = \{\gamma(s, t) : s \in \mathbb{S}^{d-1}\}$$

$$\partial_t \gamma = u \circ \gamma \quad |$$

